

Node.js & Asynchronous JavaScript Task Output Report

Course: Node.js & Asynchronous JS | **Platform:** L&T; EduTech | **Student:** Gagan Aditya

Task 1: Command-Line Task Logger — Node.js Setup & Execution

■ **File Name:** Task_1_Command_Line_Task_Logger.js | ■ **Command:** node Task_1_Command_Line_Task_Logger.js

```
zsh — Task_1_Command_Line_Task_Logger.js (Task 1)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_1_Command_Line_Task_Logger.js

Task Logger Started
```

Task 2: Understanding How Node.js Works & Node.js Architecture

■ **File Name:** Task_2_Architecture_and_Non_Blocking.js | ■ **Command:** node Task_2_Architecture_and_Non_Blocking.js

```
zsh — Task_2_Architecture_and_Non_Blocking.js (Task 2)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_2_Architecture_and_Non_Blocking.js

Task Logger Started
Immediate Message: Triggered fs.readFile (demonstrating non-blocking asynchronous execution)

--- [Async Callback] File Contents ---
Buy groceries
Finish Node.js CIA-2 Assignment
Review code quality
```

Task 3: NodeJS Resources & Working with NodeJS Examples

■ **File Name:** `Task_3_NodeJS_Resources_Doc_Examples.js` | ■ **Command:** `node Task_3_NodeJS_Resources_Doc_Examples.js`

```
zsh — Task_3_NodeJS_Resources_Doc_Examples.js (Task 3)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_3_NodeJS_Resources_Doc_Examples.js

Task Logger Started

--- [Official Doc Example Adaptation] tasks.txt Contents ---
Buy groceries
Finish Node.js CIA-2 Assignment
Review code quality
```

Task 4: NodeJS REPL Introduction & Date Timestamping

■ **File Name:** `Task_4_NodeJS_REPL_Integration.js` | ■ **Command:** `node Task_4_NodeJS_REPL_Integration.js`

```
zsh — Task_4_NodeJS_REPL_Integration.js (Task 4)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_4_NodeJS_REPL_Integration.js

[8/25/2026, 10:22:14 AM] Task Logger Started

[8/25/2026, 10:22:14 AM] Logged Tasks:
[8/25/2026, 10:22:14 AM] Task #1: Buy groceries
[8/25/2026, 10:22:14 AM] Task #2: Finish Node.js CIA-2 Assignment
[8/25/2026, 10:22:14 AM] Task #3: Review code quality
```

Task 5: Node Process Object, Command Line & Terminal I/O

■ **File Name:** `Task_5_Process_CommandLine_IO.js` | ■ **Command:** `echo "y" | node Task_5_Process_CommandLine_IO.js`
"Complete Task 5 Verification"

```
zsh — Task_5_Process_CommandLine_IO.js (Task 5)

gaganaditya@MacBook-Pro CIA_2_LnT % echo "y" | node Task_5_Process_CommandLine_IO.js "Complete Task 5 Verification"

Received task from CLI: "Complete Task 5 Verification"

Are you sure you want to save this task? (y/n): Success: Task "Complete Task 5 Verification" saved successfully to tasks.txt.
```

Task 6: Node Packages – NodeMon & Monitoring Applications

■ **File Name:** *Task_6_NodeMon_Monitoring.js* | ■ **Command:** *npx nodemon Task_6_NodeMon_Monitoring.js*

```
zsh — Task_6_NodeMon_Monitoring.js (Task 6)

gaganaditya@MacBook-Pro CIA_2_LnT % npx nodemon Task_6_NodeMon_Monitoring.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node Task_6_NodeMon_Monitoring.js`

=====
[10:22:46 AM] Task Logger running with Nodemon monitoring!
Edit this file and save to observe nodemon automatically restart the application.
=====
```

Task 7: Debugging Node Programs & Debugging Techniques

■ **File Name:** *Task_7_Debugging_Techniques.js* | ■ **Command:** *node Task_7_Debugging_Techniques.js*

```
zsh — Task_7_Debugging_Techniques.js (Task 7)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_7_Debugging_Techniques.js

Debugging demonstration started...
[DEBUGGED] Processing task: Fix typo in variable name
Debugging demonstration finished successfully.
```

Task 8: Asynchronous Programming & Callback Functions

■ **File Name:** Task_8_Async_Programming_Callbacks.js | ■ **Command:** node Task_8_Async_Programming_Callbacks.js

```
zsh — Task_8_Async_Programming_Callbacks.js (Task 8)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_8_Async_Programming_Callbacks.js

Calling saveTaskCallback...
Callback Success Handler: Task "Task 8: Learn Callback Patterns" appended successfully!
```

Task 9: Node Timers & Global Objects (setTimeout & setInterval)

■ **File Name:** Task_9_Node_Timers_and_Global_Objects.js | ■ **Command:** node Task_9_Node_Timers_and_Global_Objects.js

```
zsh — Task_9_Node_Timers_and_Global_Objects.js (Task 9)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_9_Node_Timers_and_Global_Objects.js

[10:24:19 AM] Task Logger Timer Service Started.
[10:24:22 AM] □ Task counter update: 1 task status check(s) completed so far.
[10:24:24 AM] □ Reminder: review your tasks!
[10:24:25 AM] □ Task counter update: 2 task status check(s) completed so far.
[10:24:28 AM] □ Task counter update: 3 task status check(s) completed so far.
[10:24:31 AM] □ Task counter update: 4 task status check(s) completed so far.
[10:24:34 AM] □ Stopped interval timer after 15 seconds.
```

Task 10: JavaScript Promises — Introduction & Detail

■ **File Name:** `Task_10_JavaScript_Promises.js` | ■ **Command:** `node Task_10_JavaScript_Promises.js`

```
zsh — Task_10_JavaScript_Promises.js (Task 10)

gaganaditya@MacBook-Pro CIA_2_LnT % node Task_10_JavaScript_Promises.js

Executing saveTaskPromise with .then() and .catch(...)
Promise Resolved [,.then()]: Task "Task 10: Master JavaScript Promises" saved successfully via Promise!
```

Task 11: Try/Catch Error Handling & Async-Await Concepts

■ **File Name:** `Task_11_Try_Catch_and_Async_Await.js` | ■ **Command:** `node Task_11_Try_Catch_and_Async_Await.js`

```
gaganaditya@MacBook-Pro CIA_2_LnT % node Task_11_Try_Catch_and_Async_Await.js
```

```
=== 1. Valid File Path Test ===
```

```
[Async/Await] Attempting to save task to "tasks.txt"...
```

```
SUCCESS [try block]: Task "Task 11: Test Async/Await Success" saved to tasks.txt
```

```
=== 2. Non-existent Folder Path Test (Fires Catch Block) ===
```

```
[Async/Await] Attempting to save task to "./non_existent_folder/invalid_tasks.txt"...
```

```
FAILURE [catch block fired]: Failed to save task!
```

```
  Error Details: ENOENT: no such file or directory, open './non_existent_folder/invalid_tasks.txt'
```

```
  Error Code    : ENOENT
```